AlphaBlend: Hardware-Algorithm Co-design with Mixed-Alphabet Set Multipliers for DNN Workloads

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***Abstract*—We introduce Mixed Alphabet-Aware Training (Mixed AAT), a novel hardware-aware training method that assigns distinct low-complexity multiplier sets (alphabet sets) to individual Deep Neural Network (DNN) layers based on sensitivity to weight perturbations. This method maintains full-precision accuracy while significantly reducing memory and compute costs. Achieving <1% accuracy degradation on ResNet, VGG, and MobileNet across CIFAR-10 and ImageNet, we complement Mixed AAT with two hardware solutions: an approximate near-memory digital architecture and a modified systolic array accelerator. These co-designs - a blend of Mixed AAT and hardware solutions - offer at least 3.5X energy savings and 1.3X speedup, ideal for energy-constrained applications.**

I. Introduction

DNNs are essential for image classification, detection, and language translation tasks. However, their Matrix-Vector Multiplications (MVMs) consume substantial energy and memory, limiting deployment in energy-constrained devices [1]. To address this, we propose a hardware-aware training approach using Alphabet Set Multipliers (ASMs) [2], [3]. ASMs use 4-bit integer "alphabets" to approximate weight-input multiplication (refer Section II). Eight alphabets can cover the full range of 4-bit integer weights/inputs, constituting the full alphabet set for 4-bit integer representation. We introduce approximation in ASMs by reducing the number of alphabets in the alphabet set. This reduces the compute intensive MVMs into low-complexity multiplications involving mostly shift and add operations. Even though reducing the alphabet set saves energy, it impacts accuracy significantly. While methods in [2] apply a fixed alphabet set across all DNN layers, yielding good results on smaller datasets, they suffer significant accuracy loss on larger models and complex datasets like ImageNet. Our proposed *Mixed Alphabet-Aware Training (AAT)* alleviates such accuracy issues by optimizing alphabet selection to balance energy efficiency and accuracy. Also, we propose novel hardware solutions that will provide substantial energy savings and compute efficiency compared to the general ASM-accelerator integration provided in [2].

In Mixed AAT, different alphabet sets are assigned to individual DNN layers based on their sensitivity to weight perturbations, calculated using Taylor series-based sensitivity analysis (detailed in Section III). This approach maintains full-precision accuracy while reducing energy and computational demands. We demonstrate these gains through a systolic array architecture with ASM-enhanced processing elements and introduce a near-memory digital hardware architecture that reduces data movement and storage for further energy efficiency. By blending approximate ASMs with general-purpose and optimized hardware architectures, our method delivers energy-efficient DNN deployment without compromising performance, making it ideal for energy-constrained environments. We propose the following key contributions:

- **Mixed AAT**: Our method mitigates accuracy loss from ASM approximation by assigning customized alphabet sets to DNN layers via Taylor-based sensitivity analysis, achieving *near zero accuracy loss* on CIFAR-10 and ImageNet across MobileNet, ResNet, and VGG models for image classification tasks.
- **Near-multiplier-less MVM implementations**: The systolic array achieves 5X energy savings and 2X latency reduction by blending ASMs into its processing elements. The near-memory implementation further improves energy efficiency to 7X and latency to 4.5X by reducing storage by around 50% (customized for AAT) and minimizing communication overhead. These results are based on applying a single alphabet set across all DNN layers. Using a mixed-alphabet set provides at least *3.5X energy savings and a 1.3X speedup*, without performance loss.

II. Alphabet Set Multipliers (ASM)

Background: ASMs decompose DNN weights into smaller bit sequences, each represented by unique 4-bit segments referred to as *alphabets*. Table I shows that a 4-bit weight segment can be efficiently expressed as an addition of shifted versions (shift and add) of a small set of unique 4-bit sequences, such as 0001 (1), 0011 (3), and 0101 (5). These sequences, or *alphabets*, form a reduced set of numerical representations used to approximate the weights. The complete alphabet set includes eight unique sequences: {1, 3, 5, 7, 9, 11, 13, 15}. This approach reduces the computational complexity of multiplication in MVM units by replacing multiplications with shifts and additions using alphabet representations.

Selection of Optimal Alphabet Set: Limiting the number of alphabets reduces both computational and memory overhead with minimal accuracy loss. To implement approximation, alphabets are prioritized based on their ability to generate multiple 4-bit numbers: {1, 3} > {5, 7} > {9, 11, 13,

TABLE I
WEIGHT DECOMPOSITION INTO SHIFTED VERSIONS OF ALPHABETS

Number of Weight Bits	Weights	Weight Decomposition
8	W1 = 1000 0110	$W1 = 2^7(0001) + 2^1(0011)$
4	W2 = 1010	$W2 = 2^1(0101)$

15}, optimizing efficiency. High-sensitivity DNN layers are assigned complex alphabet sets (e.g., $\{1, 3, 5, 7\}$) to minimize accuracy loss, while low-sensitivity layers use simpler sets (e.g., $\{1, 3\}$) for energy savings. As per Table II, VGG13 maintains high accuracy on CIFAR-10 (small dataset) even when using fixed reduced alphabet sets across all layers.

TABLE II
CIFAR-10 ACCURACY FOR VGG13 WITH DIFFERENT ALPHABET SETS

Alphabet Set	Accuracy (%)	Accuracy Degradation (%)
Full Precision (32-bit)	88.86	0
$\{1, 3, 5, 7\}$	88.84	0.02
$\{1, 3, 7\}$	88.81	0.05
$\{1, 3, 5\}$	88.82	0.04
$\{1, 3\}$	88.78	0.08
$\{1\}$	88.74	0.12

Single Alphabet-Aware Training: Single AAT mitigates accuracy loss from approximate ASMs using a fixed alphabet set for MVMs, maintaining performance through Quantization Aware Training (QAT). Forward passes utilize reduced alphabet sets while backward passes remain full precision. As per Table III, accuracy degradation is around 2%. To further enhance training stability, we introduce minor optimizations like a tailored learning rate schedule and custom regularization, reducing sensitivity in early layers. Our approach incorporates step-by-step quantization, starting from full precision, followed by 4-bit quantization of weights, then activations, and finally approximate ASM bits. At each step, the model adjusts to the new representation. This contrasts with Sarwar et al. [2], who directly apply ASM approximations, resulting in higher accuracy loss. By applying these step-by-step quantizations, we preserve accuracy, particularly in Mixed AAT.

TABLE III
ACCURACY RESULTS FOR SINGLE AAT

Model	CIFAR-10 Accuracy		ImageNet Accuracy			
	Baseline (%)	AAT (%)	Top-1 Baseline (%)	Top-5 Baseline (%)	Top-1 AAT (%)	Top-5 AAT (%)
ResNet50	94.33	93.90	75.756	92.678	72.60	90.702
ResNet18	94.29	93.71	69.806	89.340	67.966	87.766
MobileNetV2	94.00	93.36	65.258	86.764	62.846	84.646

III. MIXED ALPHABET-AWARE TRAINING (MIXED AAT)

Mixed AAT assigns distinct alphabet sets to DNN layers based on their sensitivity to weight changes. This allows more precise tuning of layer precision, delivering a highly efficient, low-power DNN framework without sacrificing performance.

Sensitivity Analysis for Alphabet Set Selection: The core of Mixed AAT is a Taylor expansion-based sensitivity analysis, which evaluates each DNN layer's sensitivity to weight changes. The analysis involves two components:

i) First-order term (Gradient): We compute the gradient of the loss function, $\nabla_\theta L(\theta)$, with respect to the parameters θ, capturing the immediate effect of small weight changes on the overall loss.

ii) Second-order term (Hessian): The Hessian matrix $H_L(\theta)$, representing second-order derivatives, provides insights into how the curvature of the loss landscape affects weight sensitivity. Since full Hessian computation is costly, we approximate diagonal elements or simplified variants to balance accuracy.

By expanding the loss function $L(\theta)$ around the current parameter values θ using a Taylor series, we approximate the effect of small changes in the weights:

$$L(\theta + \Delta\theta) \approx L(\theta) + \nabla_\theta L(\theta) \cdot \Delta\theta + \frac{1}{2}\Delta\theta^\top H_L(\theta)\Delta\theta$$

This expansion allows us to evaluate how sensitive each layer is to weight changes, directly influencing the alphabet set assigned during training. Once layer sensitivity is computed, layers are grouped into:

- **High Sensitivity**: Assigned complex sets (e.g., $\{1,3,5\}$) to preserve accuracy.
- **Medium Sensitivity**: Assigned intermediate sets (e.g., $\{1,3\}$) balancing energy and accuracy.
- **Low Sensitivity**: Assigned simple sets (e.g., $\{1\}$), maximizing energy efficiency.

This adaptive selection minimizes accuracy loss while ensuring significant energy savings.

We validate Mixed AAT on CIFAR-10 and ImageNet benchmarks. Mixed AAT achieves $< 1\%$ accuracy degradation compared to full-precision 32-bit models (Table IV), performing comparably to HAWQ [4], a state-of-the-art mixed-precision method. Unlike HAWQ, which uses higher bit-widths for sensitive layers, Mixed AAT maintains a uniform bit-width (4 bits) across layers, but assigns different alphabet sets.

TABLE IV
IMAGENET ACCURACY RESULTS FOR RESNET-50 WITH MIXED AAT

Quantization Type	Method	Accuracy Top-1(%)	Gap(%)
32-bit floating-point	Forward and Back-prop	75.76	-
Mixed-precision	HAWQ [4]	75.48	0.28
4-bit mixed-alphabet	Mixed AAT	75.37	0.39

IV. HARDWARE SOLUTIONS FOR AAT

A. Systolic Array Solution

We blend Mixed AAT with a systolic array by integrating ASMs into custom processing elements (PEs). For sets like $\{1\}$, we use a multiplierless dataflow (Fig 2(a)), reducing storage by 50%. For $\{1, 3\}$, a low-complexity multiplier is implemented, as shown in Fig. 1(c). Compared to general-purpose PEs (e.g., Eyeriss [5]), our ASM-blended PEs improve area and power by 2X with similar latency and bit-widths.

B. Near-Memory Custom Architecture Solution

Fig 2 illustrates the all-digital implementation of an ASM-blended Near-Memory Processing unit, using 6T SRAM as the memory bitcell. Each output node in a DNN layer is computed as $Out_j = \sum(W_{i,j} * Im_i)$, where $i = 1, \ldots, m$ and $j = 1, \ldots, n$ represent input and output nodes. The 7-bit output (O) from multiplying a 4-bit input (I) with a 4-bit decoded weight (W) is passed through an adder tree for the final result. As detailed in Section II, each multiplication is a product of the shifted versions of alphabets and inputs.

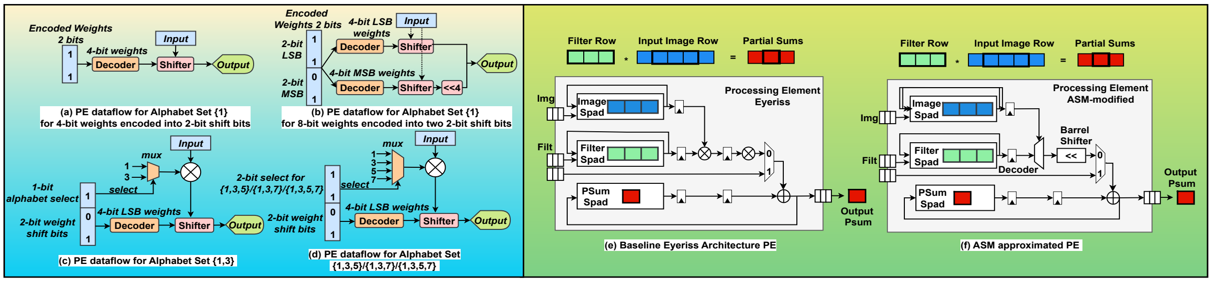

Fig. 1. Approximate ASM blended PE in DNN accelerators (Eyeriss). Notations for (c), (d): Filt- Weight filters data, Img- Image Data, Psums- Partial Sums

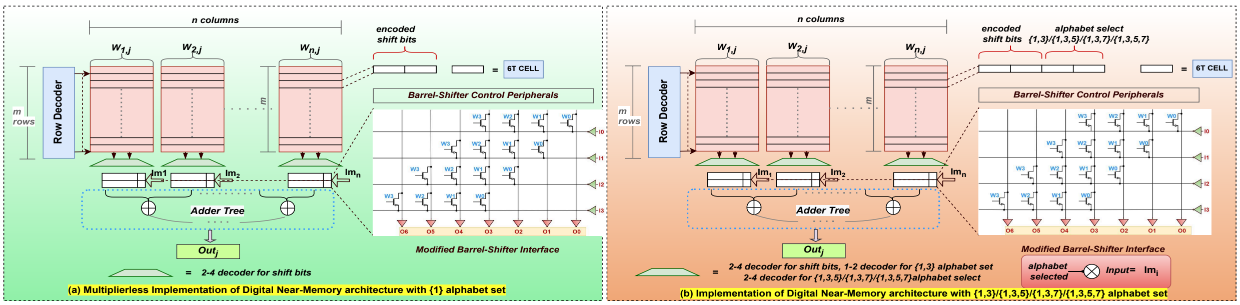

Fig. 2. Approximate ASM integrated Digital Near-Memory Processing Unit. ($W_{i,j}$ = encoded weights, W_{0-3} = bits of 4-bit decoded $W_{i,j}$, Im_i = inputs, I_{0-3} = bits of 4-bit Im, Out_j = MAC output, O_{0-6} = bits of one $Im_i \times W_{i,j}$ output) (a) Memory and peripheral design for $\{1\}$ alphabet set (b) Memory and peripheral design for $\{1,3\}/\{1,3,5\}/\{1,3,7\}/\{1,3,5,7\}$ alphabet sets

In a 4-bit precision hardware, the possible weights resulting from using the alphabet set $\{1\}$ are $\{0001,0010,0100,1000\}$. Thus, the weights $W_{i,j}$ involve the shifting of "1" in different bit positions and can be encoded into 2-bit shift bits. The encoded ASM-approximated shift bits are stored in memory bit cells and decoded using a 2-to-4 decoder into one-hot decoded weight bits, which are then fed to modified barrel shifters. We realize a 50% reduction in memory space compared to a 4-bit memory, coupled with decreased memory access energy per weight word. Input activations are provided as values for shifting within the barrel shifters. With no bit-serial inputs, the barrel shifter inputs operate in a bit-parallel manner, yielding $W * I$ output at each barrel shifter's end. This configuration enhances throughput owing to its bit-parallel input, and renders the operations multiplierless. We obtain hardware benefits but incur accuracy losses (Table V). Next, we demonstrate multiplications using weights approximated by the alphabet set $\{1,3\}$. Weights for $\{1\}$ are $\{0001, 0010, 0100, 1000\}$, encoded into two shift bits with alphabet select "0". For $\{3\}$, weights are $\{0011, 0110, 1100\}$, requiring two shift bits for encoding the alphabet "11" and alphabet select "1". All bits in the SRAM memory storage per word will be zero for propagation of "0". This approach reduces SRAM storage by 33% compared to 4-bit weights, with less accuracy loss compared to $\{1\}$, but requires additional bits for alphabet selection and a 2-bit multiplier and 1-to-2 decoder per $W * I$. Multi-alphabet sets {1,3,5},{1,3,7}, {1,3,5,7} show no storage advantage beyond 4 bits, thus standard accelerators are preferred for such cases (Table V). For error-sensitive DNN layers, we employ multi-alphabet sets based on permissible accuracy deviations, while for less sensitive layers, we use multiplierless configurations.

TABLE V
NEAR-MEMORY CATEGORIZATION FOR ALPHABET SETS

Alphabet Set	Multiplierless	Memory Savings %	Accuracy loss % (wrt VGG-13)
{1}	Yes	50	0.12
{1,3}	No (low complexity)	33	0.08
{1,3,5}/{1,3,7}/{1,3,5,7}	No	0	<0.04

Reconfigurability: The architecture is reconfigurable, allowing the weight array to be divided into "D" partitions for increased parallel MAC outputs. This boosts peripheral units (barrel shifters, adders, accumulators) without expanding memory. For instance, setting D = 2 yields two MAC outputs per cycle. Input precision ranges from 4 to 16 bits, and weight precision scales in 4-bit increments (e.g., 4b, 8b, 16b), embedding ASM approximations, offering flexible trade-offs between accuracy and energy efficiency.

C. Experimental Results

Experimental Setup: We implemented the near-memory processing unit and ASM-integrated PE in TSMC 65nm CMOS for 4b/4b precision of weights and input activations. Energy, area, and latency metrics were measured using Cadence Virtuoso and Synopsys Design Compiler (DC), with HSPICE simulations under TT process and at 25°C. Monte-Carlo simulations with 3-sigma variation and 1000 seeds confirmed no errors at 0.9V. For system-level analysis, we used the Timeloop-Accelergy framework [6], [7], which maps DNN workloads, providing layer-wise energy and area estimates. We benchmarked VGG workloads against the Eyeriss architecture, and modifications to the Timeloop-Accelergy PiM tool

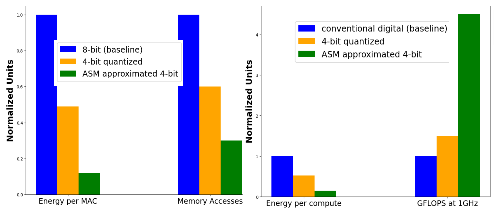

Fig. 3. (a) Energy and Memory Access Comparisons per MAC for 128x256 SRAM array at iso-latency (b) Energy and Throughput comparisons between ASM blended Digital Near Memory Processing Unit and conventional DNN accelerator architecture for VGG workloads

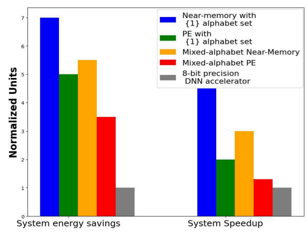

Fig. 4. Comparison of our different hardware solutions against 8-bit digital DNN accelerator for VGG13 model

captured the system-level aspects of the digital near-memory architecture.

Results for Systolic Array Solution: ASM blended PE achieves *3X power savings and 3X memory savings* compared to 8-bit baseline PE in an Eyeriss DNN accelerator architecture at iso-latency. The clock period is set at 10ns with 50% duty cycle with maximum data activity rate. For system results, we obtain operation counts and area for VGG13 workloads using the Timeloop-Accelergy tool. Energy per compute and energy per MAC are evaluated for all VGG layers in the Eyeriss architecture, comprising 14*12 PEs arranged spatially. Table VI shows that with ASM integrated PEs, replacing multipliers with decoders and shifters, we achieved 5X energy savings and 2X system speedup due to reduced memory storage. Note that these results are for the fixed alphabet set $\{1\}$.

TABLE VI
SYSTEM ENERGY COMPARISON BETWEEN ASM-BLENDED PES AND GENERAL DNN PES FOR VGG WORKLOADS

PE type	Energy per compute	Energy savings Ratio	GFLOPS @1GHz
Eyeriss PE (baseline)	93.53pJ	1X	3928
ASM based PE (ours)	18.82pJ	5X	7842

Results for Near-Memory Hardware Solution: We achieve $\sim 7X$ system energy savings and $\sim 4.5X$ system latency improvement (Fig 3(b)) using a cycle-accurate accelerator as the baseline. Each layer operates on a 128x128 SRAM array, with energy per MAC and memory access details shown in Fig 3(a). These results apply to the fixed alphabet set $\{1\}$.

V. State of the Art Results

We benchmark our *Mixed AAT* method against SOTA methods such as *DeepShift*, *INQ*, *HAWQ*, and traditional 8-bit DNN accelerators. Table VII shows that our Mixed AAT achieves competitive results, balancing energy efficiency and accuracy. For VGG-13 models, our method achieves zero accuracy loss.

TABLE VII
IMAGENET ACCURACY RESULTS AGAINST SOTA
W: WEIGHT BITS, MP = MIXED-PRECISION, MA = MIXED-ALPHABET

Model	Method	W	Top-1 (%)	
			Accuracy	Gap
ResNet50	Baseline	32	75.75	-
	DeepShift [8]	5	75.93	-0.18
	INQ [9]	5	75.01	0.74
	HAWQ [4]	MP	75.48	0.28
	Single AAT (ours)	4	73.6	2.15
	Mixed AAT (ours)	MA(4)	75.37	0.39

Our approach utilizes *Mixed alphabet sets* to optimize energy consumption with competitive accuracy across all tested models. *Fig 4* illustrates that ASM-blended near-memory and systolic array solutions outperform 8-bit digital DNN accelerators for *VGG13* workloads in terms of energy efficiency, achieving approximately **5X energy savings** and a **2X system speedup** compared to the Eyeriss-like baseline accelerator.

Hardware Overhead: Our systolic array PEs require minimal architectural changes, focusing primarily on updating the processing elements (PEs). For additional communication savings, the near-memory approach further reduces overhead by employing fewer peripherals and lower storage requirements compared to standard digital near-memory architectures. We replace expensive multipliers with simpler shift-add units and store shared alphabet sets, reducing complexity.

Discussion: Mixed AAT surpasses methods like DeepShift and INQ by offering layer-specific alphabet sets, enabling more approximation in less sensitive layers without significant accuracy loss. While DeepShift and INQ rely on non-uniform quantization or binary shifts, Mixed AAT can implement both strategies through customized alphabet sets. Moreover, Mixed AAT demonstrates hardware efficiency, unlike DeepShift and INQ, which lack hardware-focused evaluations. Compared to HAWQ, which adjusts bit-widths using Hessian-based analysis, Mixed AAT integrates seamlessly into hardware through ASMs. Additionally, the constant bit-width in Mixed AAT simplifies hardware integration across all DNN layers.

VI. Conclusion

Our work presents Mixed Alphabet Aware Training, a hardware-aware training methodology that assigns different alphabet sets to each DNN layer through Taylor series based sensitivity analysis. This approach delivers significant energy efficiency (3.5X energy savings and 1.3X speedup) over conventional accelerators by blending low-complexity ASMs in two hardware solutions- systolic-array and customized Near-Memory hardware architecture. Our co-design solutions effectively limits accuracy degradation due to ASM approximations to less than 1% on CIFAR-10 and ImageNet models, making it highly suitable for energy-efficient DNN deployments across edge and cloud environments. Future research will focus on expanding Mixed AAT to other architectures such as transformers and refining the hardware design for optimized performance in edge computing applications.

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